Toward Accelerated LLM Inference: Porting and Evaluating Diffusion-Based Speculative Decoding on TPU

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Problem & Motivation

- LLM decoding is **inherently sequential**: generating n tokens requires n forward passes through the full target model, making latency grow linearly with output length.
- **Speculative decoding** uses a lightweight *draft model* to propose K candidates; the target verifies all in one batched forward pass. Higher-quality drafts yield more accepted tokens per step.
- **Our contribution:** Port DFlash—a diffusion-based block drafter—from GPU/PyTorch to TPU/JAX inside `tpu-inference`. Benchmark Qwen3-4B across 9 datasets (math, code, chat).
- **Why TPU?** Verification cost is **flat from K=16 to K=128** (0.97×), while GPU scales to 2.3×—ideal for aggressive speculative decoding with wide draft blocks.

Background

Speculative Decoding

- A fast draft model generates K candidates; the target scores all K in one pass via rejection sampling. Accepted tokens are emitted *without recomputation.*
- Key metric: τ = average accepted tokens per draft step. Ideal $\tau \to K$; limited by draft–target distribution mismatch.

Draft Model Approaches

- **N-gram:** Near-zero latency but low τ (poor approximation).
- **Eagle3:** Autoregressive drafter; high τ but $O(K)$ sequential passes—draft latency grows with block size.
- **DFlash:** Diffusion-based block drafter—all K=16 tokens in **one pass** via non-causal attention. Draft cost is $O(1)$ regardless of K, uniquely suited for TPU's flat verification profile.

DFlash Architecture

- 4-layer transformer with non-causal block attention; each draft token attends bidirectionally to all other draft tokens.
- Reuses target model's embedding layer and LM head (no extra params).
- Conditioned on target hidden states from layers $[1, 9, 17, 25, 33]$, projected via FC + RMSNorm.
- Trained on only 289K samples; outperforms Eagle3 (1.4M samples).

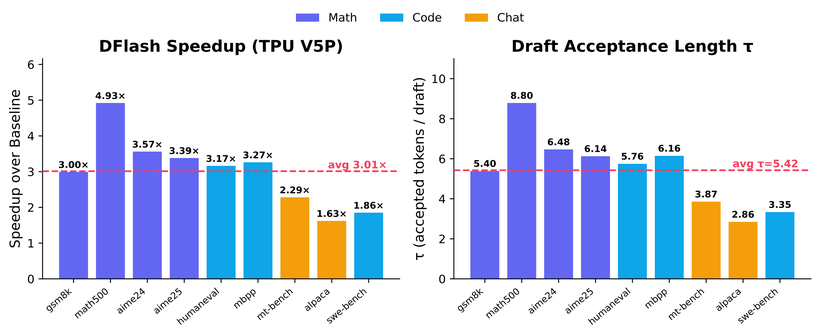
Methods: GPU → TPU

1. **Dual KV cache:** Paged KV (PagedAttention) for target; static JAX KV with `dynamic_update_slice` for draft.
2. **Non-causal attention:** TPU `flash_attention` with `causal=False` for bidirectional draft-token attention.
3. **Critical bug fix—sequence-length inflation:** Phantom tokens (∼15/step) corrupted RoPE embeddings. Fixing this *nearly doubled performance:*
   τ: 2.49 → 4.48 Speedup: 1.30× → 2.31×
4. **Hidden-state extraction:** Layers [1,9,17,25,33] concatenated and projected via learned FC + RMSNorm.
5. **Method registration:** New `"dflash"` branch in `tpu_runner.py` with prewarm and cache-reset support.

Zero new vLLM dependencies. Full DFlash contract preserved.

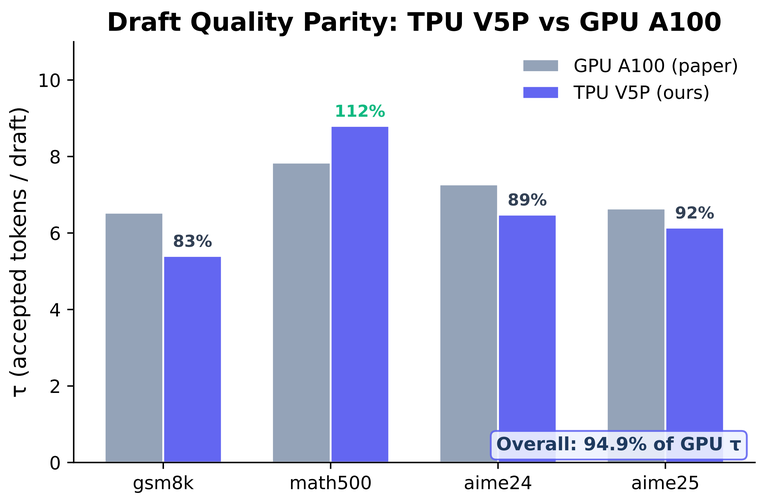
Key Results

3.01× Standalone **τ = 5.42** Accepted/Draft **2.31×** Pipeline



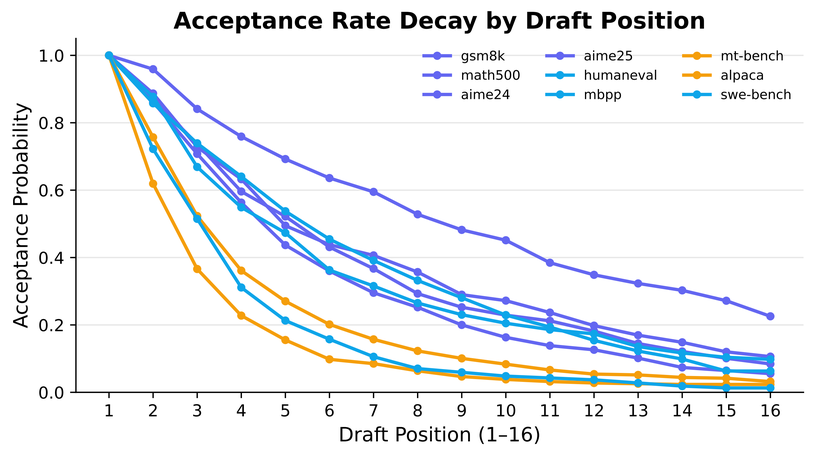
- **Setup:** Qwen3-4B + DFlash block-16 on TPU V5P (4 chips), evaluated on **9 benchmarks** in 3 categories.
- **Math** (gsm8k, math500, aime24, aime25): **3.72**× avg. speedup, τ=6.71. *Math500 peaks at 4.93*× (τ=8.80), exceeding GPU paper's τ=7.84.
- **Code** (humaneval, mbpp): **2.77**× avg. speedup, τ=5.09. Structured patterns boost acceptance.
- **Chat** (mt-bench, alpaca, swe-bench): **1.96**× avg. speedup, τ=3.37. Higher entropy limits draft accuracy.
- **Output quality preserved:** Divergence is numerical (bf16 rounding), not semantic errors.

TPU vs GPU Draft Quality



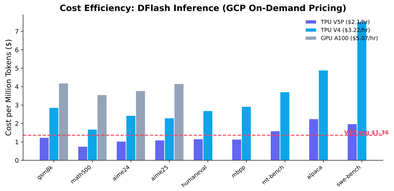
- **Draft quality parity:** Our TPU port achieves **94.9%** of GPU A100 τ across the 4 math benchmarks with available GPU data.
- Math500 *exceeds* GPU at **112%**—the only benchmark where TPU surpasses the original GPU implementation.
- Remaining ∼5% gap attributed to bf16 precision and minor attention kernel divergence between PyTorch and JAX.

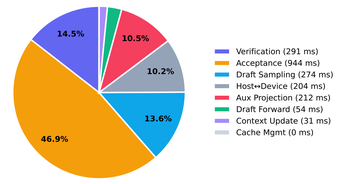
Acceptance Rate by Position



- Math benchmarks maintain >70% acceptance through positions 8–10, reflecting high predictability of step-by-step reasoning.
- Chat tasks drop below 50% by position 4–5 due to higher output entropy and creative language patterns.
- DFlash's **non-causal attention** lets each position attend bidirectionally, sustaining quality at later draft positions.
- Even at position 16, math retains >20% acceptance, justifying wide K=16 blocks.

Cost Efficiency & Profiling





- TPU V5P: **$1.36/M tokens** avg. vs. $2.08 (V4) and $2.94 (GPU A100 est.)—the most cost-efficient option.
- Verification (46.9%) and acceptance (14.5%) dominate step time; draft forward is only 2.7% (∼54 ms).
- **Pipeline overhead** (host–device transfer, cache mgmt.) explains the standalone–pipeline τ gap (5.42 vs 4.48).

Conclusion & Future Work

- DFlash on TPU V5P: **3.01× standalone**, **2.31× pipeline** speedup with no quality loss.
- TPU's flat verification scaling makes it the ideal hardware for speculative decoding with large draft blocks.
- Most cost-efficient inference configuration among all hardware–method combinations tested.

Future Work

- **Wider blocks (K=64, 128):** Both draft and verify stay flat on TPU—should scale near-linearly.
- **vLLM optimization:** Reduce host–device overhead to close the standalone–pipeline gap.
- **LM head fusion:** Two matmuls account for ∼30% of step time and can be fused into one kernel.

References & Acknowledgements

References

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